SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items *(2)*	Comparison Depth(2.5)	Use of Evidence *(2)*	Critical Thinking *(2)*	Clarity of Presentation *(1)*	***Total(10)***
Weightage	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I B.Lakshmi Lohitha(192324182) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:B.Lakshmi Lohitha

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer	Income (₹)	Spending Score
C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:
Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

Product	**Sales (₹)**	**Frequency**
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:
Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5 . Dataset: Mixed Attributes

Item	**Price (₹)**	**Category**	**Rating**
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:
Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

CO4-AT3

Name:- B. lakshmi lohitha
RegNo:- 192324482

Q1) K-Means Vs Hieararchial clustering

⇒ K-means clustering divides the customers into a fixed no. of clusters using centroids.

⇒ It is fast and works well for large datasets.

⇒ It repetedly assigns customers to the nearst centroid and updates the centroid.

⇒ Hierarchial clustering creates clusters step-by-step

⇒ It uses a dendrogram to show relationships b/w customers.

⇒ It is slower and requires more computation for large datasets

conclusion:-

K-means is better for large datasets customer data, while hierarchial clustering is useful for understanding cluster relationships.

Q2) K-Means Vs EM Algorithms

⇒ K-means assigns each document to only one cluster.

⇒ It works better when cluster & are clearly separeted

→ Em gives a probability of a document belonging to different clusters.

⇒ There fore, a document containing words from two topics can have membership in both clusters.

⇒ this is useful to the given document data because some documents may have overlapping word patterns.

conclusion :-

em is better than K-means for handling over-lapping document clusters.

Q3) Geographic Data- Euclidean vs Manhatton

Given,

P1-P5 have latitude values from 12.97 - 13.10 and longitude values from 80.20 - 80.30

• Ecludiean distance measures the direct straight time distance b/w two geographic points:

Formula -

$$d = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$$

• manhattan distance adds the absolute latitude and longitude differences.

- Formula

$$d = |x_1 - x_2| + |y_1 - y_2|$$

Conclusion:-

Euclidean distance is more suitable here because the data represents geographic locations and direct spatial closeness is important.

Q4) Retail products - Agglomerative vs Divisive

Given sales ⇒ 20,000 → ₹ 52,000 and frequency → -22

- Agglomerative clustering starts with each product as a separate cluster.
- It gradually merges similar products based on sales and frequency.
- For example, P2(₹50,000, 22) and P4(₹52,000, 12) have high sales and can be considered relatively similar.
- Divisive clustering starts with all five products in one cluster.
- It then splits the products into smaller groups.
- Agglomerative clustering is simple and products suitable for grouping these retail products based on sales and purchase frequency.

Q5) Mixed Attributes - centroid - Based Vs Density Based

Given : price → ₹ 250 - ₹ 550,

Category → A/B,

Rating - 3.5 - 4.5

⇒ The dataset contains numerical attributes, such as prices and Rating.

⇒ It also contains a categorical attributes, category (A/B)

⇒ centroid based clustering, such as K-means forms groups around a central point.

→ K-means mainly works with numerical values, so category must be properly encoded

⇒ Density-based clustering such as DBSCAN, forms clusters based on dense groups of products.

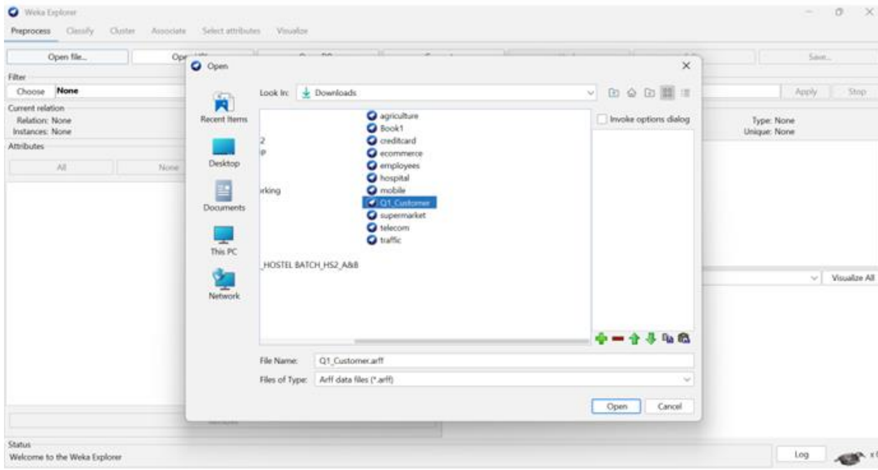
⇒ It can also identify unusual products as outliers.

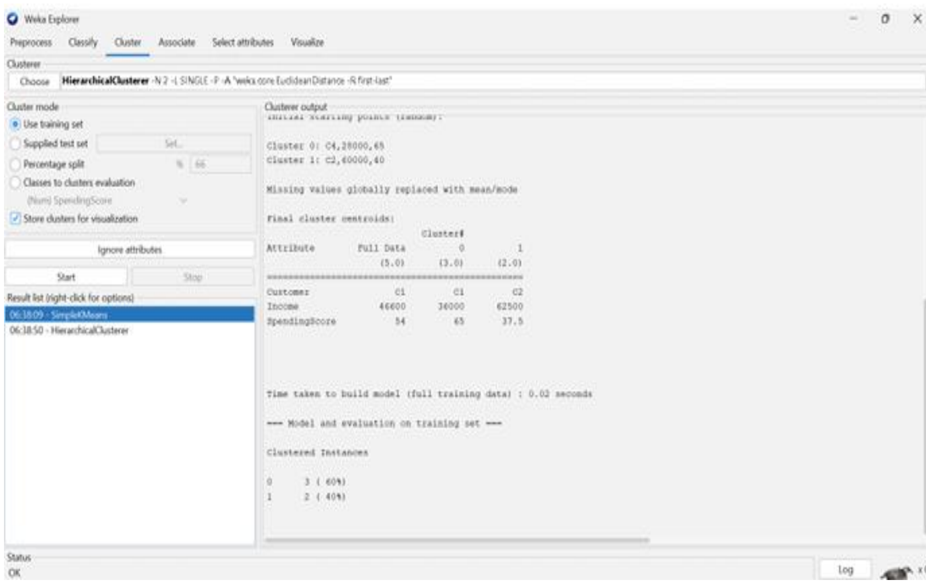
→ For this mixed dataset, suitable encoding and distance measurements are required.

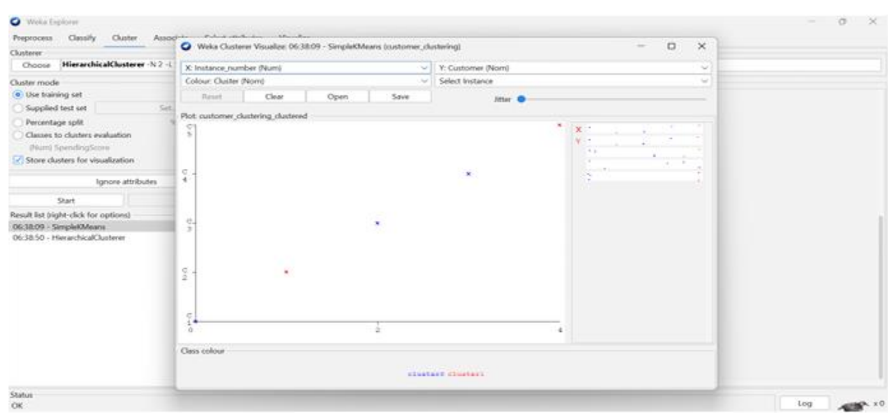
Conclusion :-

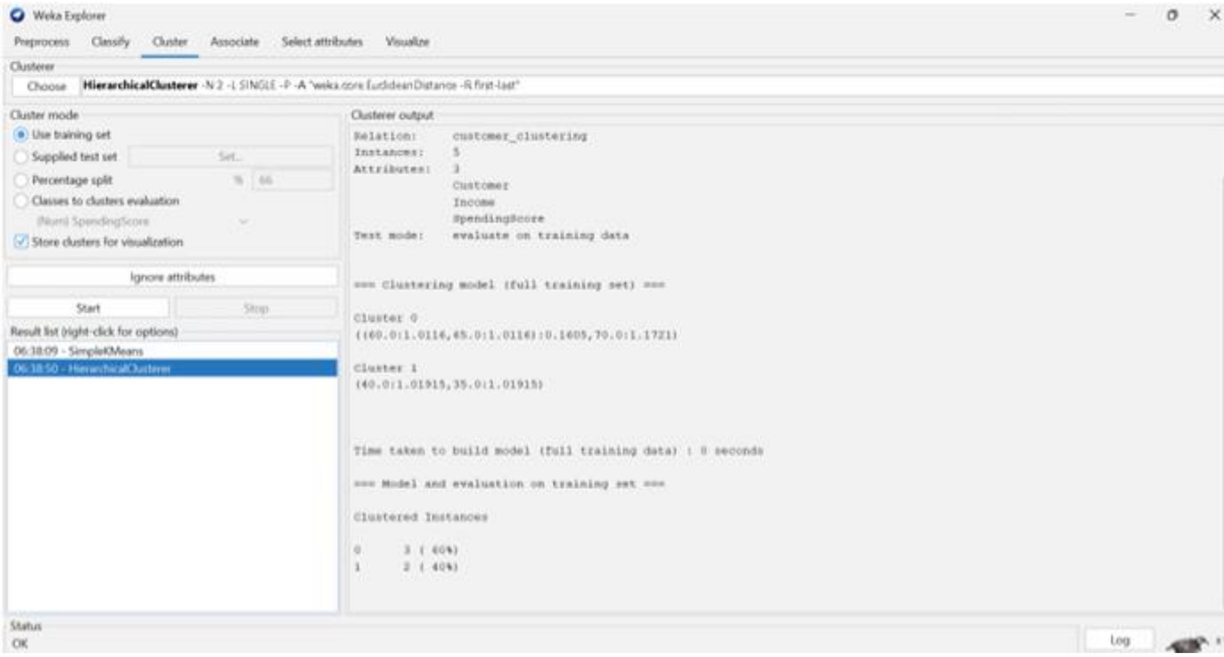
centroid - based clustering is simple for numerical attributes, while density based clustering is useful when the products form irregular groups (or) contain outliers.

1)

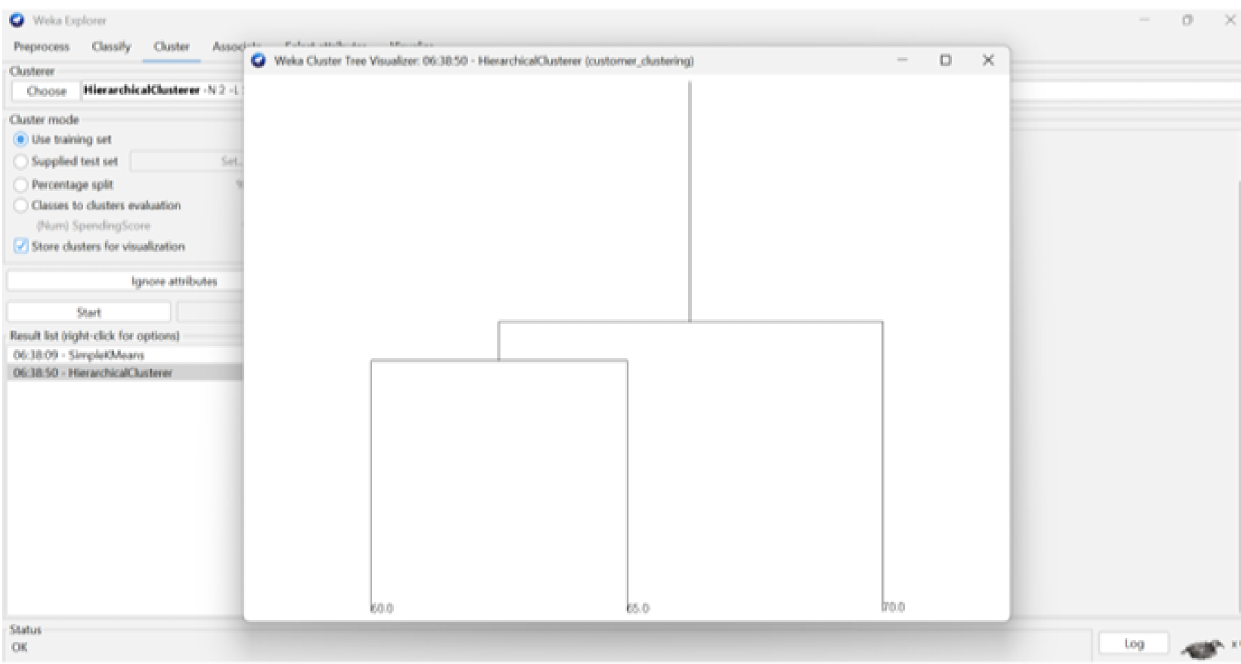

Open
Look In: Downloads
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
Q1_Customer
supermarket
telecom
traffic
File Name: Q1_Customer.arff
Files of Type: Arff data files (*.arff)
Open
Cancel


Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Choose HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:38:09 - SimpleKMeans
06:38:50 - HierarchicalClusterer
Clusterer output
Cluster 0: C4,28000,65
Cluster 1: C2,60000,40
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (3.0) (2.0)
Customer C1 C1 C2
Income 46600 36000 62500
SpendingScore 54 65 37.5
Time taken to build model (full training data) : 0.02 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Status
OK
Log

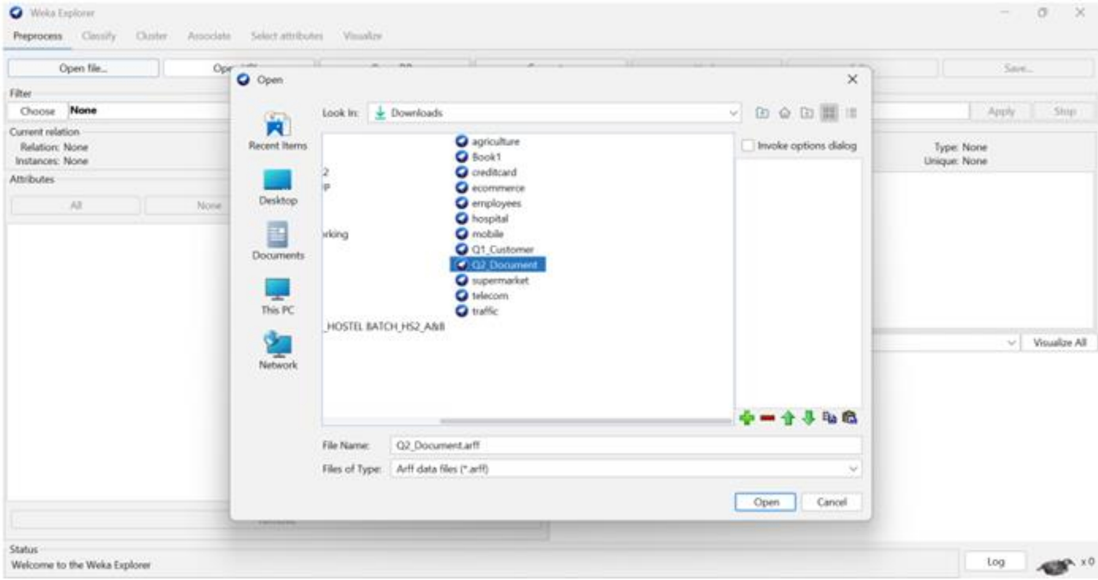

Weka Clusterer Visualize: 06:38:09 - SimpleKMeans (customer_clustering)
X: Instance_number (Num)
Y: Customer (Nom)
Colour: Cluster (Nom)
Select Instance
Reset Clear Open Save
Jitter
Plot: customer_clustering_clustered
Class colour
cluster0 cluster1



Weka Explorer
Preprocess Classify Cluster Associate Select attributes Visualize
Clusterer
Choose HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) SpendingScore
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:38:09 - SimpleKMeans
06:38:50 - HierarchicalClusterer
Clusterer output
Relation: customer_clustering
Instances: 5
Attributes: 3
Customer
Income
SpendingScore
Test mode: evaluate on training data
=== Clustering model (full training set) ===
Cluster 0
((60.0:1.0116,65.0:1.0116):0.1605,70.0:1.1721)
Cluster 1
(40.0:1.01915,35.0:1.01915)
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Status
OK
Log

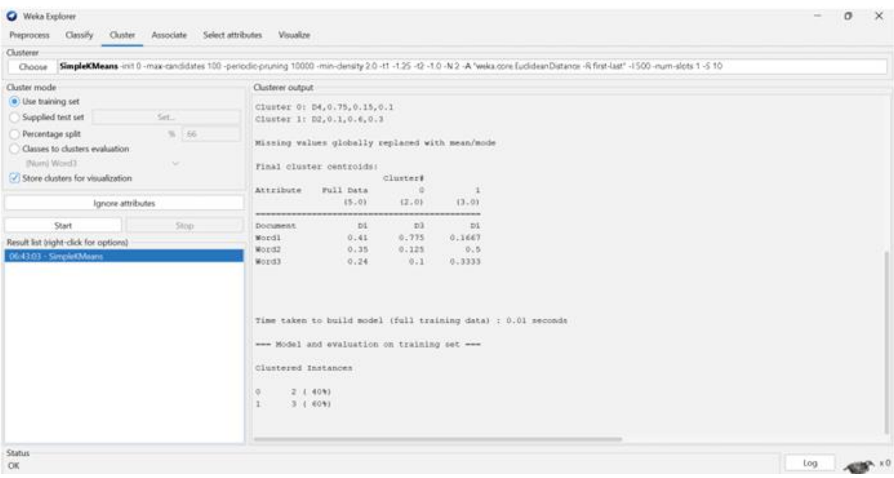


Weka Explorer
Preprocess Classify Cluster
Clusterer
Choose HierarchicalClusterer -N 2 -L
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) SpendingScore
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:38:09 - SimpleKMeans
06:38:50 - HierarchicalClusterer
Weka Cluster Tree Visualizer: 06:38:50 - HierarchicalClusterer (customer_clustering)
60.0
65.0
70.0
Status
OK
Log

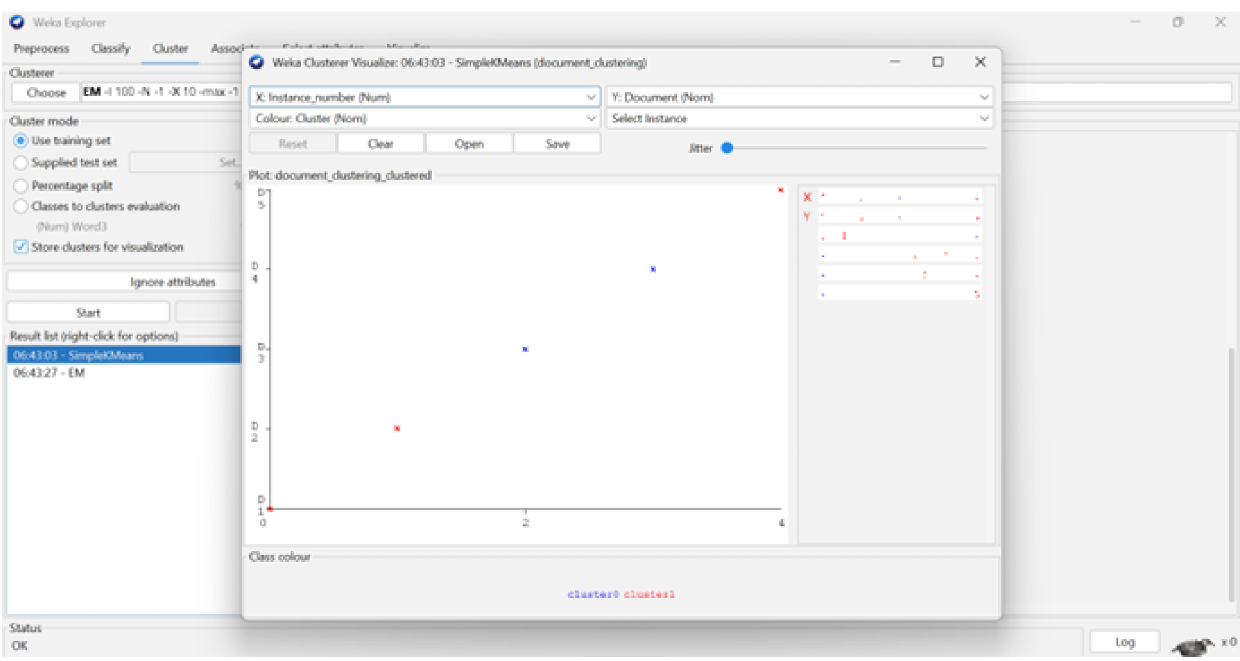


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open
Look in: Downloads
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
Q1_Customer
Q2_Document
supermarket
telecom
traffic
Invoke options dialog
File Name: Q2_Document.arff
Files of Type: Arff data files (*.arff)
Open
Cancel
Status
Welcome to the Weka Explorer

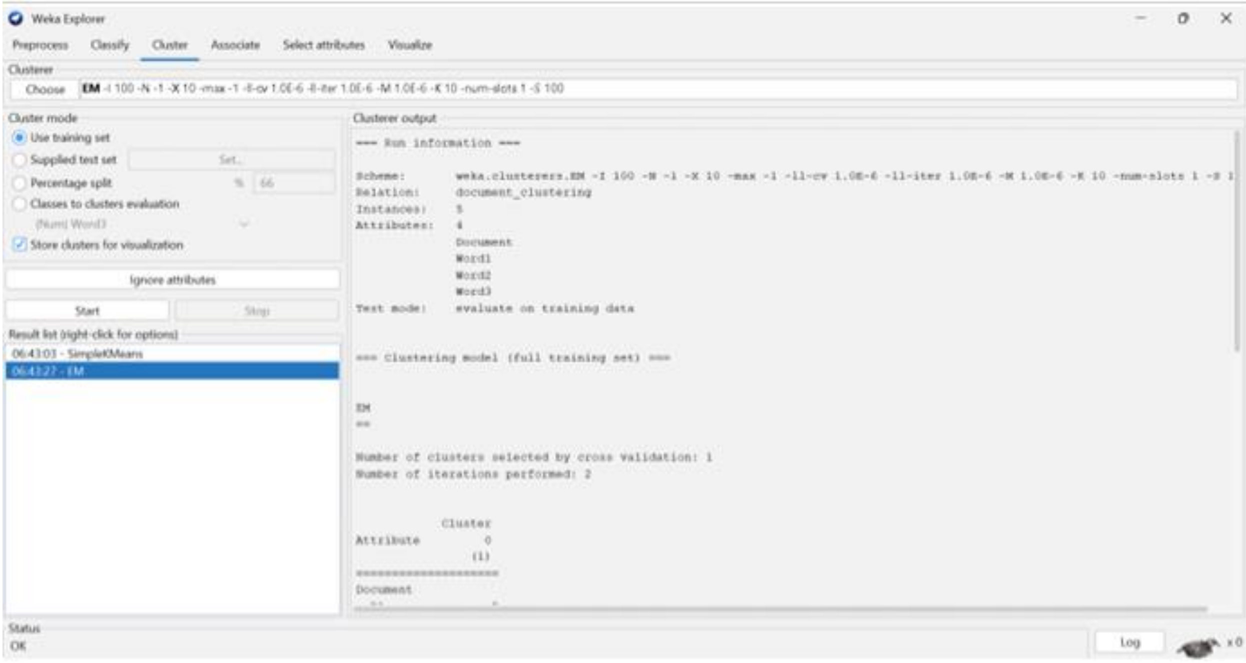
2)

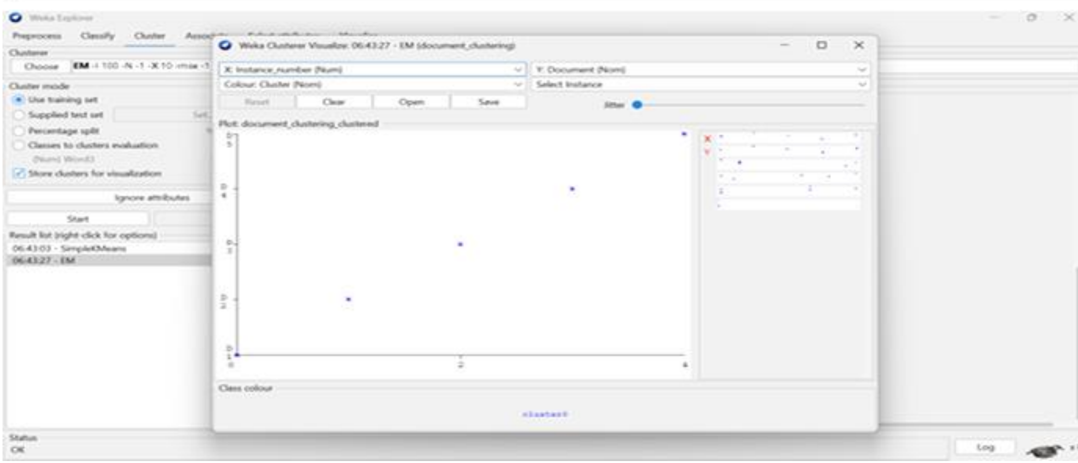


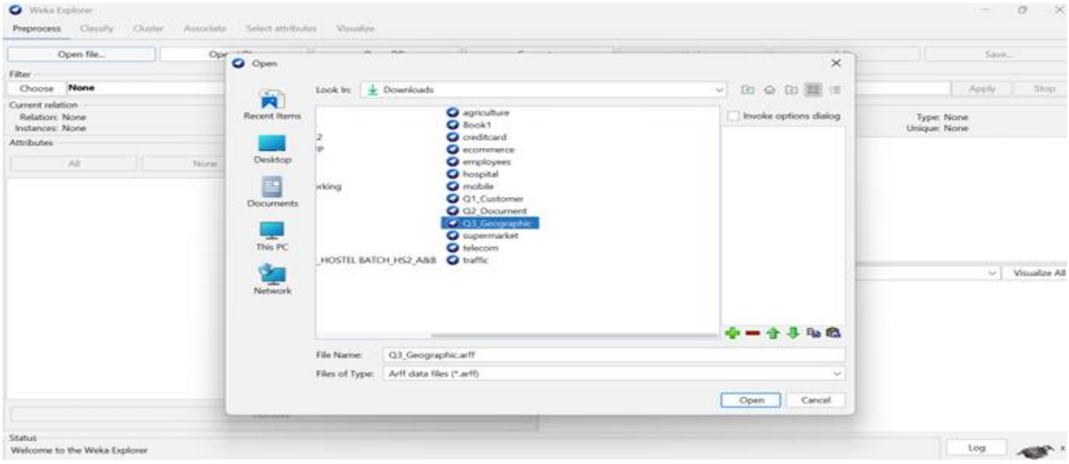
Weka Explorer
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:43:03 - SimpleKMeans
Clusterer output
Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Document D1 D3 D1
Word1 0.41 0.775 0.1667
Word2 0.35 0.125 0.5
Word3 0.24 0.1 0.3333
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log

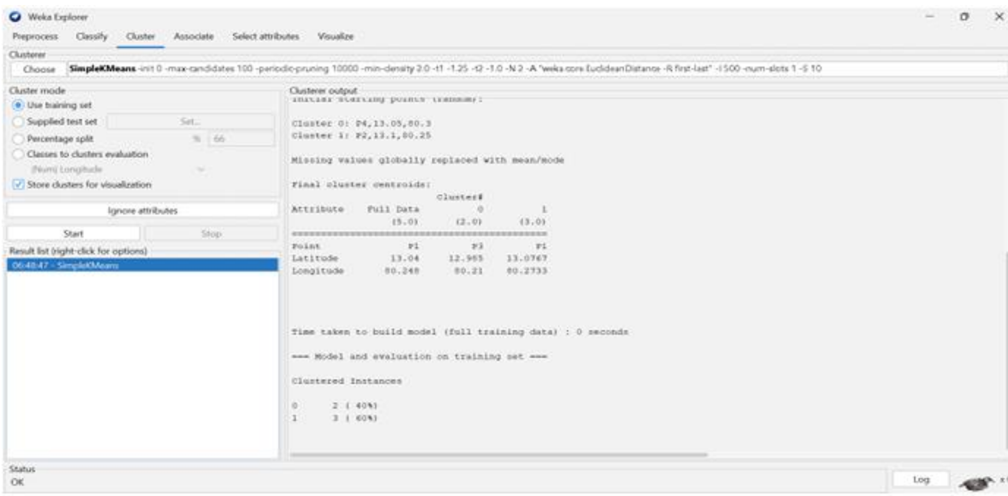


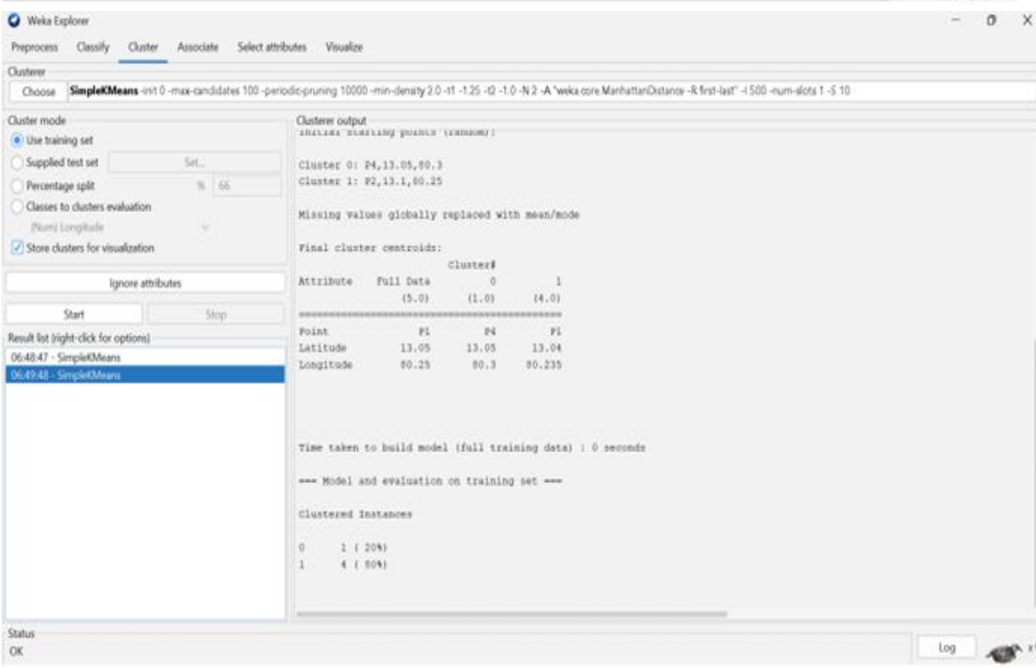
Weka Explorer
Clusterer
EM -I 100 -N -1 -X 10 -max -1
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Word3
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Weka Clusterer Visualize: 06:43:03 - SimpleKMeans (document_clustering)
X: Instance_number (Num)
Y: Document (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: document_clustering_clustered
Class colour
cluster0 cluster1
Status
OK
Log


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:43:03 - SimpleKMeans
06:43:27 - EM
Clusterer output
=== Run information ===
Scheme: weka.clusterers.EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 1
Relation: document_clustering
Instances: 5
Attributes: 4
Document
Word1
Word2
Word3
Test mode: evaluate on training data
=== Clustering model (full training set) ===
EM
==
Number of clusters selected by cross validation: 1
Number of iterations performed: 2
Cluster
Attribute 0
(1)
Document
Status
OK
Log


Weka Clusterer Visualize: 06:43:27 - EM (document_clustering)
X: Instance_number (Num)
Y: Document (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: document_clustering_clustered
Class colour
cluster0

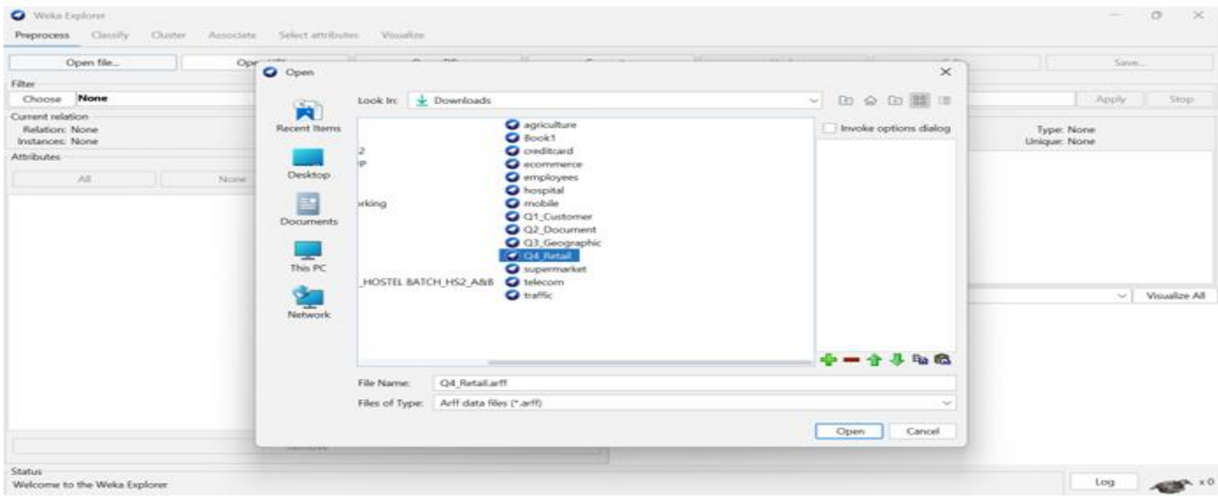

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open
Look In: Downloads
Recent Items
Desktop
Documents
This PC
Network
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
Q1_Customer
Q2_Document
Q3_Geographic
supermarket
telecom
traffic
Invoke options dialog
File Name: Q3_Geographic.arff
Files of Type: Arff data files (*.arff)
Open
Cancel
Relation: None
Instances: None
Type: None
Unique: None
Visualize All
Status
Welcome to the Weka Explorer
Log

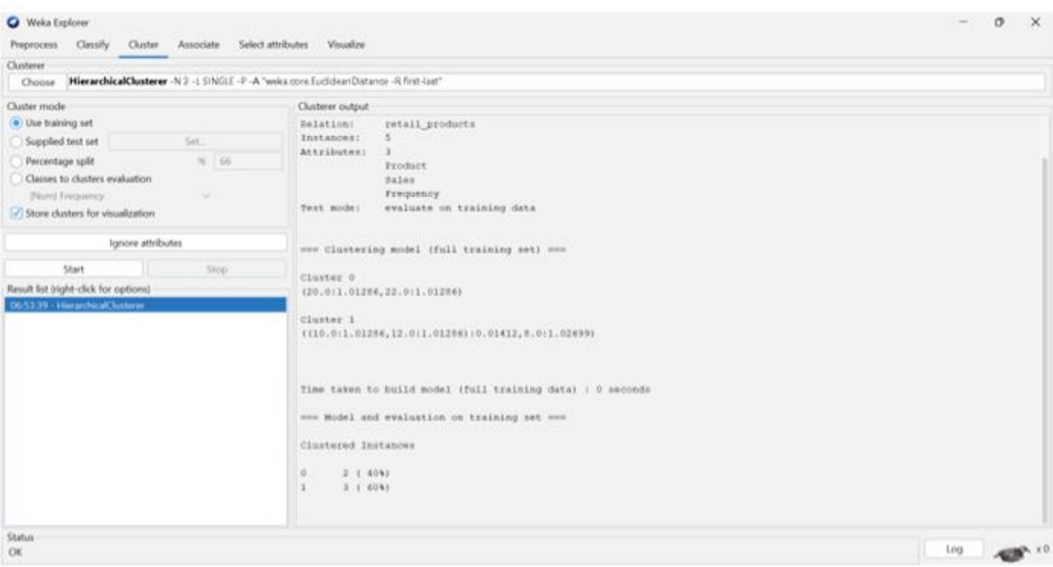

Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:48:47 - SimpleKMeans
Clusterer output
Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Point P1 P3 P1
Latitude 13.04 12.965 13.0767
Longitude 80.248 80.21 80.2733
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log


Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.ManhattanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:48:47 - SimpleKMeans
06:49:48 - SimpleKMeans
Clusterer output
Cluster 0: P4,13.05,80.3
Cluster 1: P2,13.1,80.25
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (1.0) (4.0)
Point P1 P4 P1
Latitude 13.05 13.05 13.04
Longitude 80.25 80.3 80.235
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 1 (20%)
1 4 (80%)
Status
OK
Log

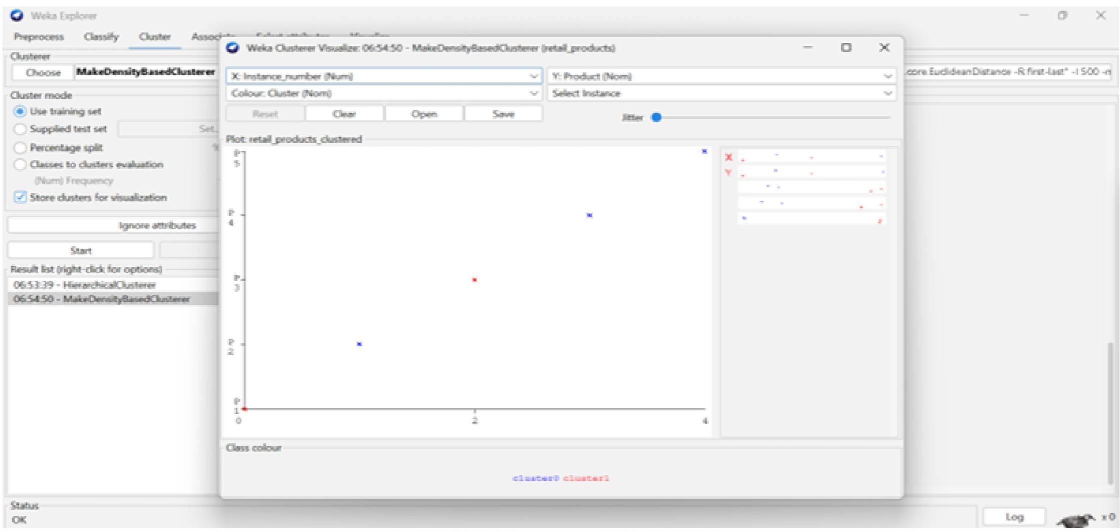

Weka Explorer
Weka Clusterer Visualize: 06:49:48 - SimpleKMeans (geographic_clustering)
X: Instance_number (Num)
Y: Point (Nom)
Colour: Cluster (Nom)
Select Instance
Plot: geographic_clustering_clustered
Class colour
cluster0 cluster1

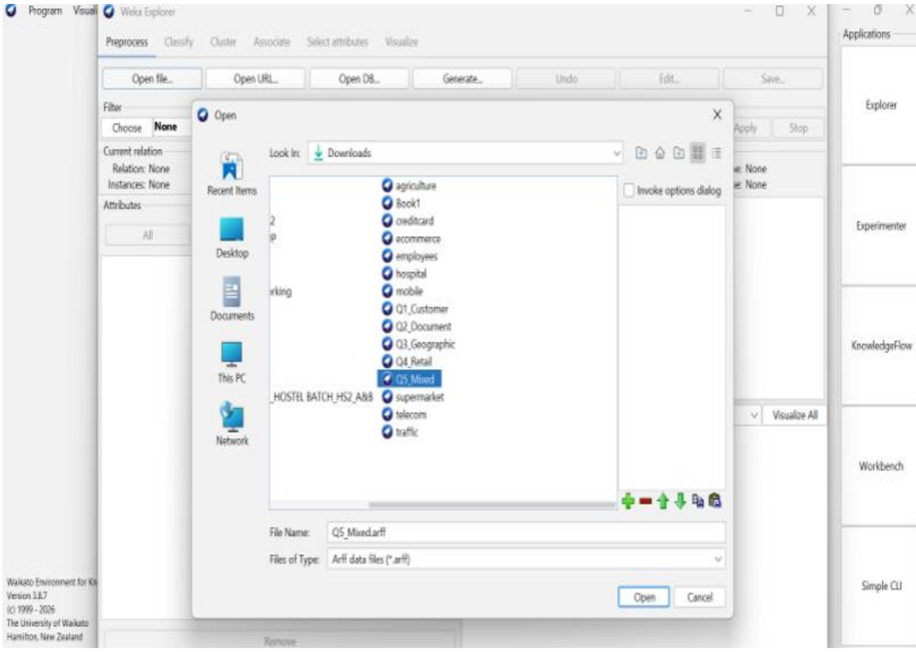
4)


Weka Explorer
Open
Look In: Downloads
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
Q1_Customer
Q2_Document
Q3_Geographic
Q4_Retail
supermarket
telecom
traffic
File Name: Q4_Retail.arff
Files of Type: Arff data files (*.arff)
Open
Cancel
Welcome to the Weka Explorer

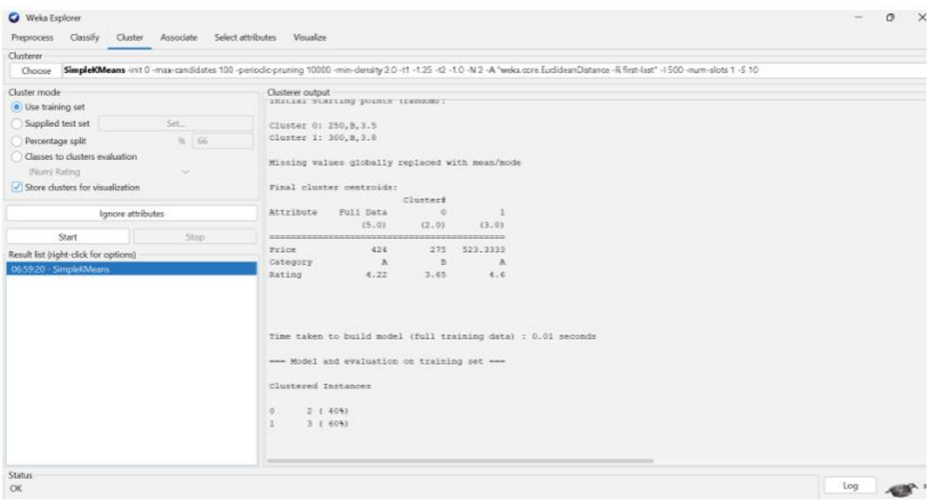

Weka Explorer
HierarchicalClusterer -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"
Clusterer output
Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency
Test mode: evaluate on training data
=== Clustering model (full training set) ===
Cluster 0
(20.0:1.01286,22.0:1.01286)
Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)


Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Clusterer
Choose
MakeDensityBasedClusterer
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:53:39 - HierarchicalClusterer
06:54:50 - MakeDensityBasedClusterer
Clusterer output
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633
Cluster: 1 Prior probability: 0.4286
Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 3 (60%)
1 2 (40%)
Log likelihood: -12.6953
Status
OK
Log

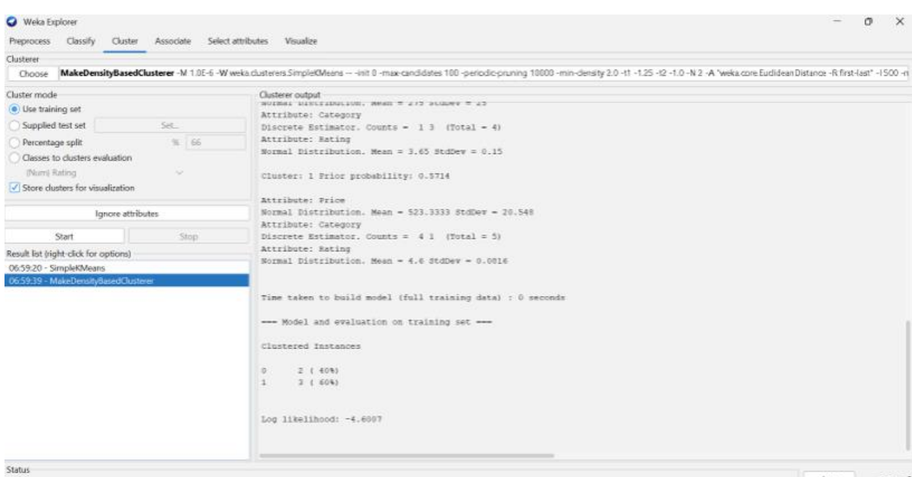

Weka Explorer
Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)
X: Instance_number (Num)
Y: Product (Nom)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: retail_products_clustered
Class colour
cluster0 cluster1
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:53:39 - HierarchicalClusterer
06:54:50 - MakeDensityBasedClusterer
Status
OK
Log



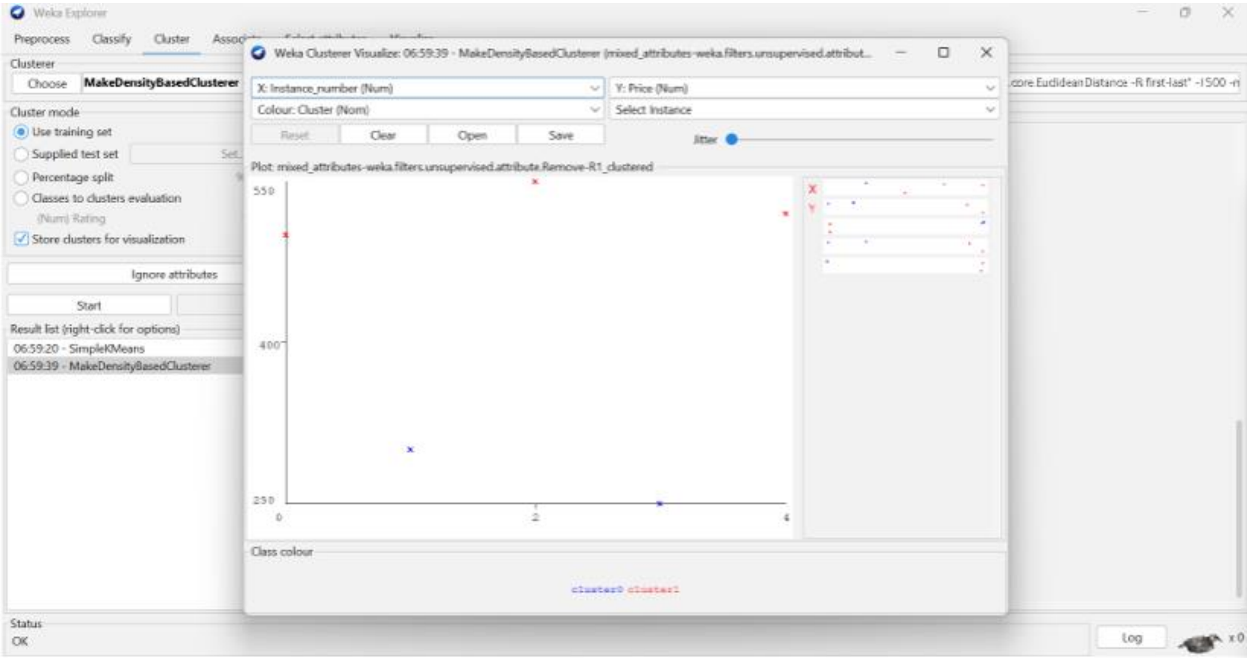
Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Open
Look In: Downloads
Recent Items
Desktop
Documents
This PC
Network
agriculture
Book1
creditcard
ecommerce
employees
hospital
mobile
Q1_Customer
Q2_Document
Q3_Geographic
Q4_Retail
Q5_Mixed
supermarket
telecom
traffic
Invoke options dialog
File Name: Q5_Mixed.arff
Files of Type: Arff data files (*.arff)
Open
Cancel
Applications
Explorer
Experimenter
KnowledgeFlow
Workbench
Simple CLI
Version 3.8.7
(c) 1999 - 2026
The University of Waikato
Hamilton, New Zealand



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Choose
SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:59:20 - SimpleKMeans
Clusterer output
Cluster 0: 250,B,3.5
Cluster 1: 300,B,3.8
Missing values globally replaced with mean/mode
Final cluster centroids:
Cluster#
Attribute Full Data 0 1
(5.0) (2.0) (3.0)
Price 424 275 523.3333
Category A B A
Rating 4.22 3.65 4.6
Time taken to build model (full training data) : 0.01 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Status
OK
Log



Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Choose
MakeDensityBasedClusterer -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
Store clusters for visualization
Ignore attributes
Start
Stop
Result list (right-click for options)
06:59:20 - SimpleKMeans
06:59:39 - MakeDensityBasedClusterer
Clusterer output
Attribute: Category
Discrete Estimator. Counts = 1 3 (Total = 4)
Attribute: Rating
Normal Distribution. Mean = 3.65 StdDev = 0.15
Cluster: 1 Prior probability: 0.5714
Attribute: Price
Normal Distribution. Mean = 523.3333 StdDev = 20.548
Attribute: Category
Discrete Estimator. Counts = 4 1 (Total = 5)
Attribute: Rating
Normal Distribution. Mean = 4.6 StdDev = 0.0816
Time taken to build model (full training data) : 0 seconds
=== Model and evaluation on training set ===
Clustered Instances
0 2 (40%)
1 3 (60%)
Log likelihood: -4.6007
Status
OK
Log


Weka Explorer
Preprocess
Classify
Cluster
Clusterer
Choose
MakeDensityBasedClusterer
Cluster mode
Use training set
Supplied test set
Percentage split
Classes to clusters evaluation
(Num) Rating
Store clusters for visualization
Ignore attributes
Start
Result list (right-click for options)
06:59:20 - SimpleKMeans
06:59:39 - MakeDensityBasedClusterer
Status
OK
Log
Weka Clusterer Visualize: 06:59:39 - MakeDensityBasedClusterer (mixed_attributes-weka.filters.unsupervised.attribut...
X: Instance_number (Num)
Y: Price (Num)
Colour: Cluster (Nom)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot: mixed_attributes-weka.filters.unsupervised.attribute.Remove-R1_clustered
550
400
250
0
2
4
Class colour
cluster0 cluster1